

Discussion notes 2

ECE 269

February 2024

1 Span is a subspace

Given a set of vectors, $V_1, V_2, V_3, \dots, V_K \in \mathcal{V}$, where $\mathcal{V}$ is a vector space over the field $\mathcal{F}$. Then the span of $V_1, V_2, V_3, \dots, V_K$ is defined as,

$$\text{Span}\{V_1, V_2, V_3, \dots, V_K\} = \left\{ \sum_{i=1}^K \alpha_i \cdot V_i, \quad \alpha_i \in \mathcal{F} \right\}$$

Hence, span of $V_1, V_2, V_3, \dots, V_K$ is essentially the set of all the vectors that can be written as the linear combination of $V_1, V_2, V_3, \dots, V_K$.

We now show that this set indeed forms a subspace. From lecture, we need to show the following,

- If $U, V \in \text{Span}\{V_1, V_2, V_3, \dots, V_K\}$, then $U+V \in \text{Span}\{V_1, V_2, V_3, \dots, V_K\}$.

Since $U, V \in \text{Span}\{V_1, V_2, V_3, \dots, V_K\}$, they can be written as a linear combination of $V_1, V_2, V_3, \dots, V_K$, i.e.,

$$U = \sum_{i=1}^K \alpha_i \cdot V_i, \quad \alpha_i \in \mathcal{F}, \forall i = 1, \dots, K$$

$$V = \sum_{i=1}^K \beta_i \cdot V_i, \quad \beta_i \in \mathcal{F}, \forall i = 1, \dots, K$$

$$U + V = \sum_{i=1}^K \alpha_i \cdot V_i + \sum_{i=1}^K \beta_i \cdot V_i$$

Using commutativity of vector addition and distributivity property of vector space.

$$U + V = \sum_{i=1}^K (\alpha_i + \beta_i) \cdot V_i$$

Since α_i and $\beta_i \in \mathcal{F}$, $\alpha_i + \beta_i \in \mathcal{F}$, $\forall i = 1 \dots K$, using the closure under addition property of field. Thus,

$$U + V = \sum_{i=1}^K (\alpha_i + \beta_i) \cdot V_i \in \text{Span}\{V_1, V_2, V_3, \dots, V_K\}$$

- If $U \in \text{Span}\{V_1, V_2, V_3, \dots, V_K\}$, then $\gamma \cdot U \in \text{Span}\{V_1, V_2, V_3, \dots, V_K\}$ $\forall \gamma \in \mathcal{F}$.

Since $U \in \text{Span}\{V_1, V_2, V_3, \dots, V_K\}$, U can be written as a linear combination of $V_1, V_2, V_3, \dots, V_K$ as above,

$$U = \sum_{i=1}^K \alpha_i \cdot V_i, \quad \alpha_i \in \mathcal{F}, \forall i = 1, \dots, K$$

$$\gamma \cdot U = \gamma \cdot \left(\sum_{i=1}^K \alpha_i \cdot V_i \right)$$

Using the distributive property of vector space,

$$\gamma \cdot U = \sum_{i=1}^K \gamma \cdot (\alpha_i \cdot V_i)$$

Using the associativity of scalar multiplication property of vector space,

$$\gamma \cdot U = \sum_{i=1}^K (\gamma \alpha_i) \cdot V_i$$

Since $\gamma, \alpha_i \in \mathcal{F}$, $\gamma \alpha_i \in \mathcal{F}$, $\forall i = 1 \dots K$, using the closure under multiplication property of field. Thus,

$$\gamma \cdot U = \sum_{i=1}^K (\gamma \alpha_i) \cdot V_i \in \text{Span}\{V_1, V_2, V_3, \dots, V_K\}$$

Thus, $\text{Span}\{V_1, V_2, V_3, \dots, V_K\}$ forms a subspace.

2 Unique representation property of any vector in the span of a set of linearly independent vectors

Let $V_1, V_2, V_3, \dots, V_K$ be “linearly independent” vectors in the vector space $\mathcal{V}$ over the field $\mathcal{F}$. We showed that the span of $V_1, V_2, V_3, \dots, V_K$ is a subspace above. Now, we will prove the unique representation property of any vector in the span of $V_1, V_2, V_3, \dots, V_K$.

Proof by contradiction: Since U is in the span of $V_1, V_2, V_3, \dots, V_K$,

$$U = \sum_{i=1}^K \alpha_i \cdot V_i, \quad \alpha_i \in \mathcal{F}, \forall i = 1, \dots, K$$

Let there be another representation of U ,

$$U = \sum_{i=1}^K \beta_i \cdot V_i, \quad \beta_i \in \mathcal{F}, \forall i = 1, \dots, K, \quad \exists \beta_l \neq \alpha_l$$

$$U = \sum_{i=1}^K \alpha_i \cdot V_i = \sum_{i=1}^K \beta_i \cdot V_i$$

$$\sum_{i=1}^K \alpha_i \cdot V_i = \sum_{i=1}^K \beta_i \cdot V_i$$

Assume, $-\beta_i$ is the additive inverse of β_i . Now add $\sum_{i=1}^K (-\beta_i)V_i$ on both sides.

$$\sum_{i=1}^K \alpha_i \cdot V_i + \sum_{i=1}^K (-\beta_i) \cdot V_i = \sum_{i=1}^K \beta_i \cdot V_i + \sum_{i=1}^K (-\beta_i) \cdot V_i$$

Using commutativity of vector addition and distributivity property of vector space.

$$\sum_{i=1}^K (\alpha_i + (-\beta_i)) \cdot V_i = \sum_{i=1}^K (\beta_i + (-\beta_i)) \cdot V_i = \sum_{i=1}^K (0) \cdot V_i = \sum_{i=1}^K \mathbf{0} = \mathbf{0}$$

From discussion-1 notes, $0 \cdot V_i = \mathbf{0} \in \mathcal{V}$, where $\mathbf{0}$ is the zero of vector space $\mathcal{V}$.

$$\sum_{i=1}^K (\alpha_i + (-\beta_i)) \cdot V_i = \mathbf{0}$$

Since $V_1, V_2, \dots, V_K$ are linearly independent vectors, their linear combination is $\mathbf{0}$, if and only if,

$$\alpha_i + (-\beta_i) = 0, \forall i = 1 \dots K$$

Adding β_i on both sides.

$$(\alpha_i + (-\beta_i)) + \beta_i = 0 + \beta_i$$

Using the associativity of addition property and addition with zero property of field.

$$\alpha_i + ((-\beta_i) + \beta_i) = \beta_i$$

Since $-\beta_i$ is the additive inverse of β_i in $\mathcal{F}$,

$$\alpha_i + (0) = \beta_i$$

$$\alpha_i = \beta_i, \forall i = 1 \dots K$$

which is a contradiction since we started with, there is at least one $\beta_l \neq \alpha_l$. Thus by contradiction, our assumption that there is another representation for U is incorrect implying that there is a unique representation for any vector in the span of a set of linearly independent vectors.

3 Relation between two sets

Given $A \in \mathcal{R}^{1 \times n}$, $A \neq 0$. How is $S_1 = \mathcal{N}(A)$ related to the set $S_2 = \{x : Ax = b, b \neq 0\}$?

We show the following,

- For any vector $v_N \in S_1$ and any vector $v \in S_2$, $v_N + v \in S_2$.

Since $v_N \in S_1$, $Av_N = 0$ and $v \in S_2$, $Av = b$. Now adding both of these,

$$Av_N + Av = 0 + b$$

$$A(v_N + v) = b$$

Thus, $v_N + v \in S_2$ for any $v_N \in S_1$ and any $v \in S_2$.

- For any vector $v \in S_2$, and an arbitrary vector $v_0 \in S_2$, $\exists v_N \in S_1$ such that $v = v_0 + v_N$.

Since $v_0 \in S_2$ and $v \in S_2$, $Av_0 = b$ and $Av = b$. Thus,

$$Av = b = Av_0$$

$$Av = Av_0$$

$$Av - Av_0 = 0$$

$$A(v - v_0) = 0$$

Say $v - v_0 = v_N$,

$$Av_N = 0$$

Observe, $v_N = v - v_0 \in \mathcal{N}(A)$. Thus, for any $v \in S_2$ and some arbitrary $v_0 \in S_2$, $\exists v_N \in S_1$, i.e. $\mathcal{N}(A)$, such that $v_N = v - v_0$, i.e. $v = v_0 + v_N$. Rewording it, any vector $v \in S_2$ can be written as a translation of $v_N \in \mathcal{N}(A)$ by $v_0 \in S_2$.